

Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data*

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Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

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